

WALID HASAN

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EDUCATION

University of Manitoba

Winnipeg, MB 2026 - 2028

MSc in Theoretical Physics

Supervisors: Prof. Andrew Frey and Prof. Margaret Carrington.

Awards: Graduate Student Research Stipend, International Graduate Student Entrance Scholarship.

BRAC University

Dhaka, Bangladesh 2022 - 2026

BS in Physics, CGPA: 3.96 (out of 4.00), Highest Distinction.

Thesis: *Warped Compactification in Supergravity: A p-Brane Model* [\[PDF\]](#)

Supervisor: Dr Ratul Mahanta and Prof. Mahbub Majumdar

Relevant Coursework: PHY401 Particle Physics, MAT313 Differential Geometry, PHY409 Black Holes Physics, PHY413 General Theory of Relativity, PHY403 Cosmology and Astrophysics, CSE482 Quantum Computing 2, MAT301 Group Theory, MAT221 Real Analysis.

Awards: 75% Merit Scholarship based on CGPA, Vice-Chancellor's and Dean's Award.

Activities:

- Took a tutorial class and prepared a problem set on manifolds for PHY413 General Theory of Relativity.
- Organised reading groups for Quantum Mechanics from Shankar's Book and QFT from David Tong and Michele Cicoli's notes.
- Wrote and presented a review paper on "Superstrings in the Early Universe" by Brandenberger and Vafa for PHY403 Cosmology and Astrophysics. [\[PDF\]](#) [\[Slides\]](#)
- Wrote and presented a review paper on "The Black Hole Information Problem" by Polchinski for PHY419 Black Holes. [\[PDF\]](#) [\[Slides\]](#)
- Presented on "Locally inertial coordinates in Kruskal" for PHY419 Black Holes. [\[Slides\]](#)

RESEARCH

Undergraduate Research Assistant at **BRAC University**

December 2025 - May 2026

PI: Professor Mahbubul Alam Majumdar

Co-PI: Dr Ratul Mahanta (USTC, Hefei) and Ahmed Rakin Kamal (Masaryk University).

Worked on a project titled "Time Dependent Backgrounds in String Theory and Early Universe Cosmology".

Received a grant of 4,000 USD from Research Seed Grant Initiative of BRAC University.

Research Intern at **ICTP PWF: Physics for Bangladesh**

July - October 2025

Supervisor: Dr Ratul Mahanta

Report: *The anatomy of non-vacuum Kasner branes* [\[PDF\]](#)

We studied time-dependent backgrounds in string theories and reviewed "Kasner Branes with Arbitrary Signature" by Wafic A. Sabra [\[2005.03953\]](#). We studied differential forms and variational techniques to derive EOM from actions and solved non-linear differential equations.

EXPERIENCE AND VOLUNTEERING

News Editor at [BracU Express](#)

March 2024 - February 2026

Wrote and edited articles for the student-run campus newspaper about student movements, animal welfare, and scientific activities on campus. List of my articles can be accessed [here](#).

Organizer at [Merul Badda Talks in Maths & Physics](#)

November 2024 -

Initiated weekly departmental seminars by inviting professors and PhD students working abroad to share their research with Bangladeshi students. Created the website and maintained outreach activities through social media.

Student Organizer at [ICTP PWF Jamal Nazrul Islam Memorial Winter School](#).

January 2025

Designed and maintained website, social media, logistics, documentation, and speaker liaison for the first residential school in theoretical physics in Bangladesh.

Mehta Fellow and Resident Counsellor at [PROMYS India](#)

May - June 2024

PROMYS is a mathematics research program for advanced high school students from South Asia. The program is supported by the Department of Mathematics at the Indian Institute of Science, Bengaluru. As a Resident Counsellor, I collaborated with faculty and a team of academic counsellors to create an inclusive and supportive environment that prioritised the health and well-being of the students.

Academic Team Member at [Bangladesh Physics Olympiad](#)

Feb 2023 - 2025

Moderated and prepared questions and marking schemes for national and regional olympiads, and coordinated training programs for IPhO.

WORKSHOP/ SCHOOL

Quantum Computing Summer School at University of Zurich (Online)

July 2025

Attended summer school focused on quantum algorithms, quantum complexity theory, quantum games, and error correction.

PWF Bangladesh: School on General Relativity and Cosmology (Online)

May - July 2025

Attended lectures and solved problem sets on advanced topics in general relativity and cosmology from leading experts.

Random Matrix Theory Summer School at Independent University, Bangladesh

June - July 2024

Taught by Dr. Arijit Chakrabarty from ISI Kolkata. Organized by Center for Computational & Data Sciences (CCDS) at IUB.

Virginia Tech Quantum Information Science and Engineering Summer School (Online)

Aug 2022

Worked with Prof. Sophia Economou and learned about the basics of Qubits, and QC circuits; used IBM's QC to run experiments.

SKILLS AND INTERESTS

Programming Language

Mathematica, Python (NumPy, pandas library), HTML, CSS, C++, L^AT_EX Bangla (Native), English (IELTS Academic Test: 8)

REFERENCES

Professor Andrew Frey
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University of Winnipeg, Canada
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